Below is a **preliminary engineering equipment list** for a **500 MW/year integrated sand-to-solar-cell polycrystalline silicon PV plant** in China.

Process route assumed:

Quartz sand → High purity quartz → MG silicon → Solar-grade polysilicon → Poly-Si ingot → Wafer → Solar cell

Technology assumptions:

- Output: **500 MW/year solar cells**
- Cell type: **Polycrystalline Al-BSF/PERC**
- Wafer thickness: 160–180 μm
- Cell efficiency: 18–20%
- Operation: 330 days/year, 24 h/day
- Chinese equipment suppliers preferred

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1. Plant Capacity Summary

Section	Annual Capacity Required
Quartz purification	5,000–8,000 ton quartz feed
MG silicon	800–1,000 ton/year
Polysilicon	3,000 ton/year
Silicon ingot	3,500 ton/year
Wafer	500 MW/year
Solar cell	500 MW/year

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2. Quartz Sand Purification Line

Purpose

Upgrade SiO_2 sand:

99% → 99.99% SiO_2

Main Equipment

Equipment	Qty	Chinese Supplier	Cost
Jaw crusher	2	Shandong Xinhai Mining Technology & Equipment	$0.3M
Ball mill	2	Henan Hongji Mining Machinery	$0.5M
Magnetic separator	3	Shandong Huate	$1M
Acid leaching reactor	4	Chinese chemical equipment makers	$2M
Washing system	1	-	$1M
Drying furnace	2	-	$1M
Purity analysis lab	1	-	$0.5M

Investment:

$5–8 million

Land:

≈ 5,000 m^2

Power:

≈ 1 MW

Water:

≈ 100–200 m^3/day

Manpower:

30–50 people

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3. Metallurgical Silicon (MG-Si) Plant

Process:

Quartz + carbon → Silicon

Purity:

98–99%

Equipment

Equipment	Qty	Supplier	Cost
Submerged arc furnace 12–15 MW	1	SMS group / Chinese suppliers	$12M
Electrode paste system	1	Chinese supplier	$2M
Transformer system	1	-	$3M
Gas cleaning	1	-	$3M
Crushing system	1	-	$2M
Dust collection	1	-	$2M

Investment:

$20–30 million

Utilities

Electricity:

≈ **14,000 kWh/ton silicon**

Annual:

≈ 12–14 GWh

Land:

10,000–15,000 m²

Water:

200–500 m^3/day

Workers:

80–120

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4. Polysilicon Plant (Most Expensive)

Technology:

Modified Siemens process

Output:

3,000 ton/year

Main Equipment

Equipment	Qty	Supplier	Cost
HCl synthesis unit	1	Chinese chemical EPC	$10M
Trichlorosilane reactor	2	-	$15M
Distillation columns	5–8	-	$25M
CVD Siemens reactors	15–20	GCL Technology technology reference	$60M
Hydrogen recovery	1	-	$10M
Chlorosilane recycle	1	-	$15M
Gas purification	1	-	$10M

Investment:

$120–160 million

Utilities:

Electricity:

80–100 kWh/kg Si

Annual:

≈250–300 GWh

Water:

1,000–2,000 m^3/day

Land:

50,000–80,000 m^2

Manpower:

300–500 people

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5. Polycrystalline Silicon Ingot Line

Technology:

Directional Solidification Furnace (DS)

Equipment

Equipment	Qty	Supplier
DS casting furnace	25–35	Jinggong Technology
Quartz crucible production	1	Chinese supplier
Argon purification system	1	-
Cutting machine	10	-

Cost:

$50–70 million

Utilities:

Power:

30–40 GWh/year

Water:

300 m^3/day

Land:

20,000 m^2

Workers:

150

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6. Wafer Production Line

Capacity:

500 MW/year

Equipment

Equipment	Qty	Supplier
Diamond wire saw	40–50	Jingsheng Mechanical & Electrical
Wire recycling system	2	Chinese supplier
Cleaning machine	5	-
Wafer sorter	5	-
Thickness tester	5	-

Cost:

$40–60 million

Utilities:

Power:

15–20 GWh/year

Water:

500–800 m^3/day

Workers:

250

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7. Solar Cell Production Line

Capacity:

500 MW/year

Equipment

Process	Machine	Supplier
Cleaning	Wet bench	Rena Technologies
Diffusion	$POCl_3$ furnace	Centrotherm International
Laser isolation	Laser system	Chinese supplier
PECVD	SiNx coating	Lead Intelligent Equipment
Screen printing	Printing line	Baccini
Firing furnace	Belt furnace	Chinese supplier

Process	Machine	Supplier
EL tester	Inspection	Chinese supplier
IV tester	Sorting	Chinese supplier

Cost:

$40–50 million

Utilities:

Electricity:

15 GWh/year

Water:

300–500 m^3/day

Workers:

300

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8. Total Factory Requirements

Land Area

Section	Area
Quartz	0.5 ha
MG silicon	1.5 ha
Polysilicon	8 ha
Ingot	2 ha
Wafer	2 ha

Section	Area
Cell	2 ha
Utilities	3 ha

Total:

20–25 hectares

(200,000–250,000 m²)

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9. Total Electricity Requirement

Approximate:

Section	GWh/year
MG silicon	12
Polysilicon	280
Ingot	40
Wafer	20
Cell	15
Utilities	20

Total:

≈380–400 GWh/year

Average power:

[
400GWh / 8760h
]

≈ **45 MW continuous load**

Recommended grid connection:

60–80 MW

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10. Water Requirement

Daily consumption:

Section	m³/day
Quartz	200
MG silicon	400
Polysilicon	1,500
Wafer	700
Cell	400
Cooling	800

Total:

≈4,000 m³/day

Install:

- RO system
- UPW system
- Wastewater treatment

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11. Manpower Requirement

Department	People
Quartz processing	50
MG silicon	120
Polysilicon	450
Ingot	150
Wafer	250
Cell	300
Maintenance	150
Quality/R&D	100
Administration	50

Total:

≈1,500 employees

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12. Total Investment Summary

Section	CAPEX
Quartz	$8M
MG silicon	$30M
Polysilicon	$150M
Ingot	$60M

Section	CAPEX
Wafer	$50M
Cell	$45M
Utilities/buildings	$70M

Total:

≈ $400–450 million

Engineering Recommendation

For a first project, a **500 MW sand-to-cell plant is technically possible but economically difficult** because polysilicon requires very large scale.

A more bankable approach:

Phase 1

- Buy polysilicon
- Build wafer + cell: 500 MW
  Investment: **$100M**

Phase 2

- Add polysilicon: 5,000–10,000 ton/year

Phase 3

- Add quartz + MG silicon

This reduces technical risk significantly.

I can next prepare a **PFD (process flow diagram), equipment layout, and detailed CAPEX/OPEX spreadsheet for this 500 MW sand-to-cell factory**.